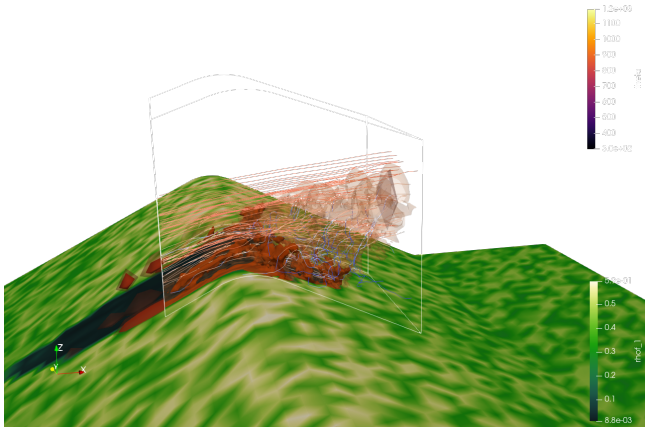


Figure 1: **[TODO: teaser caption]** Teaser image. Inside the white box: the full-resolution pass; outside it: the coarse pass.

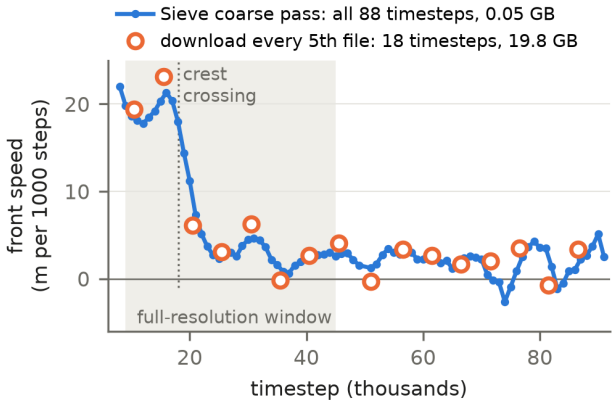


Figure 2: Front speed over the run. Blue: Sieve's coarse pass, every timestep (0.05 GB). Rings: every 5th file (19.8 GB); each ring is the speed between two consecutive files, plotted at their midpoint, from the coarse-pass data. Shaded: the full-resolution window.

4 Usage Scenarios

4.2 Tracking Wildfire Evolution with Time Windowing

4.2.1 Data and Analysis Requirements

The FIRETEC data [1] simulates a fire spreading over a ridge. It is stored on a remote server as 88 timesteps of VTK files, each 1.1 GB (96.6 GB in total) holding nine variables on a $600 \times 500 \times 61$ terrain-following grid. Once the fire crosses the crest and moves down the far slope (the lee side), its spread slows down.

Scientific Task In this scenario, the scientist asks whether at the point where the fire spread slows down significantly, the wind above the ground ahead of the fire reverses and blows against the spread. **[TODO: CITE where task is from]** Answering this question requires two kinds of data. First, the scientist must locate *when* the fire crosses the ridge and slows down. This involves sampling the series densely enough to identify the crossing and see where the speed drops; sampling density is not known in advance. Second, measuring the near-ground wind (a few meters above) requires the grid's full resolution, but only around the slowdown and near the fire.

Sieve allows the scientist to determine the relevant time window through exploration and then refine the data request around it. A spatially subsampled overview preserves all available timesteps to locate the slowdown; a subsequent request retrieves the selected window at full resolution to examine near-ground wind. Both requests are composed through Sieve's ParaView plugin [cite], combining **temporal-spatial-field** selections, and spatial subsampling as needed. This workflow avoids transferring the complete series or writing a custom server-side reduction script.

4.2.2 Coarse Pass: Locating the Slowdown

The first request, a *coarse pass*, reads every timestep, samples the grid by eight along each axis, and keeps only the five fields needed to follow the fire: the wind components (`u`, `v`, `w`), temperature (`theta`), and fuel density (`rhof_1`).

```
src = source("ssh://host//path/")
coarse = fields(src, ["rhof_1","theta","u","v","w"])
coarse = subsample(coarse, 8)
window = timesteps(src, [4000, 9000 - 45000])
window = region(fields(window, ["theta","u","v","w"]),
              x=(-100,200), y=(-120,120), z=(0,350),
              coords="world")
```

Listing 1: **[TODO: Missing caption]**

The coarse pass fetched all 88 timesteps in 47 MB. Figure 2 shows front speed falling from about 20 to a few meters per 1,000 simulation steps as the fire crosses the crest and moves down the lee side. A two-pass download spends 19.8 GB to sample every fifth file. The orange rings, calculated from the coarse-pass data at that spacing, show the broad slowdown but give speeds over 5,000-step intervals; the blue curve follows the transition at 1,000-step intervals. By subsampling space while retaining every timestep, Sieve gives the scientist this closer view of the slowdown for about 1/400 of the transfer. The shaded band marks the main interval then selected for full-resolution inspection of near-ground wind. By reducing spatial detail while retaining every timestep, Sieve lets the scientist identify the time window of interest *without* committing in advance to a sampling interval that could obscure the onset of the slowdown.

4.2.3 Full-Resolution Pass: Examining Near-Ground Wind in the Selected Window

The coarse pass shows a slightly negative wind component just as the fire crosses the crest. However, this occurs in the lowest grid layer, where wind speeds remain near zero and even small fluctuations can change the wind's direction. A sampling stride of eight retains the lowest grid layer and the next sampled layer at approximately 12 m above ground, omitting the intermediate layers.

Using the time window identified in the coarse overview, Sieve retrieved all 38 selected timesteps at full resolution, restricting the data to a box spanning the crest and lee slope and retaining only the wind components and temperature. A few meters above ground, the wind continues to blow in the direction of spread throughout the slowdown and reverses only about 16,000 simulation steps later. This later reversal therefore cannot explain the earlier slowdown. The analysis establishes *when* the wind reverses, but not *what* causes the reversal.

References

[1] R. Linn, J. Reisner, J. J. Colman, and J. Winterkamp. Studying wildfire behavior using FIRETEC. *International Journal of Wildland Fire*, 11(4):233–246, 2002. doi: 10.1071/WF02007 1

Route	Timesteps surveyed	GB moved			Bytes vs Sieve	Script lines
		coarse pass	pass 2	total		
Sieve	88	0.05	0.36	0.41	1×	6
Two-pass download	18	19.76	34.04	53.80	131×	89
Download everything	88	96.62	0	96.62	234×	73
Scripts on the server	88	0	0.61	0.61	1.5×	72

Table 1: Bytes moved by each route to the same result. The first pass surveys the series; the second fetches the window at full resolution. Sieve's requests were composed in ParaView, with no script.